

Working with IBM Granite Foundation Models!



Estimated time needed: 45 minutes

Welcome to the project, Working with IBM Granite foundation models!

In this course, you learned about IBM-built foundation models named Granite. Granite models are enterprise-level models trained on enterprise-relevant data sets from five domains: Internet, academic, code, legal, and finance.

Two variants of IBM Granite models are granite-3-3-8b-instruct and granite-8b-code-instruct. These models can be used for text generation, summarization, question and answer, classification, and extraction. In this project, you will experiment with some of the use cases for these two models.

Learning objectives:

1. Identify the use cases of IBM Granite models.
2. Experiment with IBM Granite models for different use cases, such as text summarization and question-answering.

Exercise 1: Text summarization using granite-8b-code-instruct

In this exercise, you will experiment with the **granite-8b-code-instruct** model for text summarization.

Activities

1. Create a new chat and provide a name for the chat to reflect your context, for example, Text Summarization.
2. Select the **ibm/granite-8b-code-instruct** model from the dropdown in the upper section of the right pane for this exercise.

The screenshot displays the IBM Granite AI Classroom interface. At the top, there's a header with a hamburger menu, the text "untitled chat" with an edit icon, and a "New" button. Below this is a chat window titled "Granite Code 8B" with a "Low cost" tag and "8B Parameters". The chat window contains a search bar labeled "Search models...". To the right of the chat window is a dropdown menu for selecting models. The dropdown is titled "Text Generation / Translation / Q&A" and lists several models: GPT-5 Nano (Low cost, Newest), GPT-4o Mini (Medium cost), GPT-4.1 Mini (Medium cost), Granite 3.3 8B (Low cost, 8B Parameters, Newest), and Granite 3.1 8B (Low cost, 8B Parameters). On the far right, there's a sidebar with a "Total Wait Time" of "0.00 seconds" and a message "available usage more quickly." Below this is a section titled "AI Classroom" with a large number "3" and the text "Set the prompt instructions".

3. Provide relevant instructions in the **Prompt Instructions** field, to instruct the model to read the article and generate the summary.

► [Click here for an example of a prompt instruction](#)

4. Let's ask the first question to calculate the factorial of a given number. Provide the prompt in the Type your message field. Click Start chat to generate the output.

► [Click here for an example of a prompt](#)

► [Click here for output produced by the model](#)

5. Let's ask the second question to reverse a string in Python without using built-in functions. Provide the prompt in the Type your message field to input the question. Click Send to generate the output.

► [Click here for an example of a prompt](#)

► [Click here for output produced by the model](#)

6. You can enter a third question, to count the number of vowels in a given string. Provide the prompt in the Type your message field to input the question. Click Send to generate the output. To count the number of vowels in a given string.

► [Click here for an example of a prompt](#)

► [Click here for output produced by the model](#)

Note: The output displayed is the result obtained when we entered the prompt and text. Your output may differ.

Exercise 2: Question-answering using granite-3-3-8b-instruct

Task

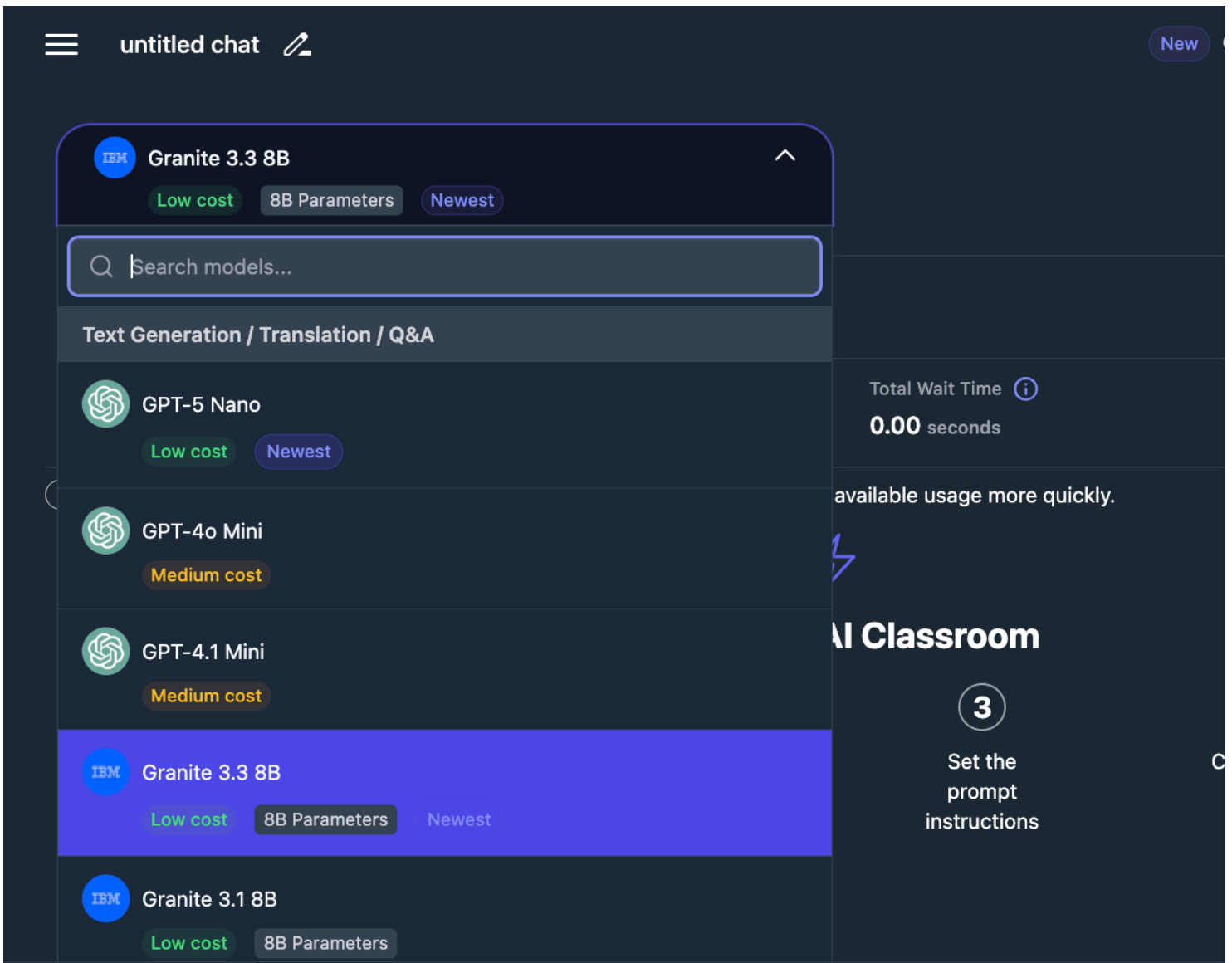
In this exercise, you will experiment with the granite-3-3-8b-instruct model for answering questions based on the following article.

Article: Tomatoes are one of the most popular plants for vegetable gardens. Tip for success: If you select varieties that are resistant to disease and pests, growing tomatoes can be quite easy. For experienced gardeners looking for a challenge, there are endless heirlooms and specialty varieties to cultivate. Tomato plants come in a range of sizes. There are varieties that stay very small, less than 12 inches, and grow well in a pot or hanging basket on a balcony or patio. Some grow into bushes that are a few feet high and wide, and can be grown in larger containers. Other varieties grow into huge bushes that are several feet wide and high in a planter or garden bed. Still, other varieties grow as long vines, six feet or more, and love to climb trellises. Tomato plants do best in full sun. You need to water tomatoes deeply and often. Using mulch prevents soil-borne disease from splashing up onto the fruit when you water. Pruning suckers and even pinching the tips will encourage the plant to put all its energy into producing fruit.

Activities

1. Create a new chat. You can name the chat according to your context.

2. Select the **ibm/granite-3-3-8b-instruct** model from the dropdown in the upper section of the right pane for this exercise.



3. Provide relevant instructions in the **Prompt Instructions** field, to instruct the model to read and answer the questions based on the given article.

► Click here for an example of an instruction

4. Let's ask the first question as a prompt. You can enter a generic question about growing tomatoes, such as 'if growing tomatoes is easy'. Provide the prompt in the **Type your message** field. Click **Start chat** to generate the output.

► Click here for an example of a prompt

► Click here for output produced by the model

5. Let's ask the second question about the varieties of tomatoes. Provide the prompt in the **Type your message** field to input the question. Click **Send** to generate the output.

► Click here for an example of a prompt

► Click here for output produced by the model

6. You can enter a third question, for example, to ask about the reason to use mulch for growing tomatoes. Provide the prompt in the **Type your message** field to input the question. Click **Send** to generate the output.

► Click here for an example of a prompt

► Click here for output produced by the model

Exercise 3: Generating answers for finance-related questions using granite-3-3-8b-instruct

Task

IBM Granite models are trained on enterprise-relevant data sets from five domains: Internet, academic, code, legal, and finance. In this exercise, you will experiment with the granite-3-3-8b-instruct model to seek responses to a few generic finance-related questions.

Activities

1. Create a new chat. You can name the chat according to your context.

2. If the **ibm/granite-3-3-8b-instruct** is not already selected in the dropdown for models, select it.

3. Let's ask the model to provide the definition of interest rates. Provide the prompt in the **Type your message** field to input the first question. Click **Start chat** to generate the output.

- ▶ [Click here for an example of a prompt](#)
- ▶ [Click here for output produced by the model](#)

5. Let's ask a follow-up question to understand how interest rates are decided. Provide the prompt in the **Type your message** field to input the third question. Click **Send** to generate the output.

- ▶ [Click here for an example of a prompt](#)
- ▶ [Click here for output produced by the model](#)

6. Let's ask another follow-up question to understand how interest rates affect savings accounts. Provide the prompt in the **Type your message** field to input the third question. Click **Send** to generate the output.

- ▶ [Click here for an example of a prompt](#)
- ▶ [Click here for output produced by the model](#)

Summary

Congratulations! You just completed the project on working with IBM Granite models. In this project, you experimented with prompts for summarization and question-answering using granite-3-3-8b-instruct and granite-8b-code-instruct models.

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